

PLCopen Help Documentation

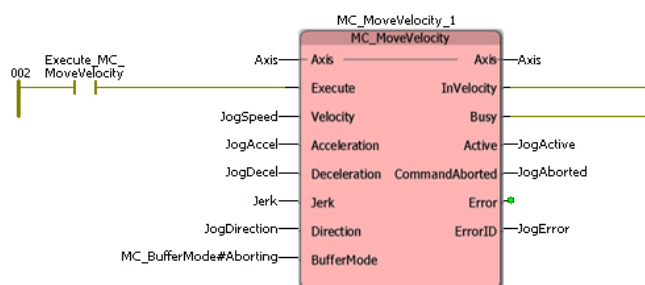
Help version created 12/1/2022

YASKAWA

Enumerated Types for PLCopen Plus

MotionWorks® IEC

Some blocks accept an enumerated type (ENUM), which is a keyword (or constant) representing a value which will configure the operation of the function block. Enumerated types are equivalent to zero-based integers (INT). Therefore, the first value equates to zero, the second to 1, etc. The format for enumerated types is as follows: ENUM:(0, 1, 2...) as displayed in the example below (MC_BufferMode#Aborting).



The following is an alphabetical listing of the Enumerated Types defined in the MotionBlockTypes file:

Enumerated Type	#INT Value	Enum Value	Description
MC_BufferMode	0	Aborting	This is the Default mode. The FB aborts an ongoing motion and the command affects the axis immediately.
	1	Buffered	The FB affects the axis as soon as the previous movement is complete. The axis will stop between the movements.
	2	BlendingLow	The FB controls the axis after the previous FB has finished, but the axis will not stop between the movements. The velocity is blended with the lowest velocity of both commands.
	3	BlendingPrevious	The FB controls the axis after the previous FB has finished (equivalent to buffered), but the axis will not stop between the movements. Blending with the velocity of the previous move.
	4	BlendingNext	The FB controls the axis after the previous FB has finished, but the axis will not stop between the movements. Blending with velocity of this (next) function.
	5	BlendingHigh	The FB controls the axis after the previous FB has finished (equivalent to buffered), but the axis will not stop between the movements. Blending with highest velocity of the previous and this (next) function.
MC_Detection_Pattern			
	0	Rising_Edge	

Enumerated Type	#INT Value	Enum Value	Description
	1	Falling_Edge	
MC_Direction			
	0	Positive_Direction	In a rotary application, forces the axis to move in a positive direction.
	1	Shortest_Way	For use in applications where the Load Type is configured as a rotary or modularized axis.
	2	Negative_Direction	In a rotary application, forces the axis to move in a negative direction.
	3	Current_Direction	For use in applications where the Load Type is configured as a rotary or modularized axis. Only applies if an existing move is in progress and another function block such as MC_MoveAbsolute or MC_MoveRelative is executed. Once the axis is at StandStill, using MC_Direction_CurrentDirection will default to the positive direction
MC_ExecutionMode			
	0	Immediately	Functionality is immediately valid. May influence ongoing motion, but not the state.
	1	Delayed	Functionality is valid when the ongoing motion command set Done, Aborted, or Error outputs.
	2	Queued	Functionality is valid when ALL previous motion commands set Done, Aborted, or Error outputs.
MC_PathChoice			
	0	ClockWise	
	1	CounterClockWise	
MC_SwitchMode			
	0	On	Unsupported
	1	Off	Unsupported
	2	EdgeOn	
	3	EdgeOff	Unsupported
	4	EdgeSwitchPositive	Unsupported
	5	EdgeSwitchNegative	Unsupported
MC_TransitionMode			
	0	TMNone	No transition curve inserted. Motion blocks are not modified. The only TransitionMode usable with the "Buffered" BufferMode.
	1	TMStartVelocity	Transition with given start velocity. Adds a segment to the motion queue unless motion is tangent.
	2	TMConstantVelocity	Transition with given constant velocity. Adds a segment to the motion queue unless motion is tangent.
	3	TMCornerDistance	Transition with given corner distance. Adds a segment to the motion queue unless motion is tangent.
	4	TMMaxCornerDeviation	Geometry limited parabolic transition with the path defined by the programmed target point to the apex of the parabola. The system will enforce this exact path, but will adjust the curve if the input segments are too short. Note that the FMK implementation is logically "ExactCornerDeviation" even though the PLCOpen spec calls for "MaxCornerDeviation"
	10	TMLXBlend	This mode is only user with MotomanSync remote hosted robots controlled by the YRC1000 series and related controllers. The transition parameter values are 0~8 with 0 defining a motion that stops at the segment end point and 1~8 apply blend distances defined in the robot controller.

Enumerated Type	#INT Value	Enum Value	Description
	11	TMMaxCornerDeviation	Geometry limited parabolic transition with the path defined by specified corner distance from the start point of the curve to the programmed target point. The blend will begin no sooner than this distance to the programmed target point and can be started at any point closer without generating an error. The system is free to pick any paths that obey this constraint. This will limit the path to a maximum distance which is useful if there are obstacles in the way. It is also useful for applications that can tolerate a variable blend radius depending on the speed, motion sequence, or timing.

Y_AdjustMode

0	MasterDistance	The cam adjustment starts immediately, and completes when the master has travelled the specified distance. If MasterDistance is 0.0, then the cam adjustment finishes in the same scan it starts.
1	ElapsedTime	The cam adjustment starts immediately, and completes within the specified time. If Time=0.0, then the adjustment completes in the same scan it starts.
2	WithinRange	The cam adjustment starts when the master is crosses the StartPosition, and completes when the master reaches the EndPosition. If the master position is already between StartPosition and EndPosition, then the adjustment starts immediately, but still completes at the EndPosition, which means that the correction speeds may be higher.

Y_ControlMode

This enumerated type is for use with Y_DirectControl

0	NoControlMode	This is not a valid value.
1	PositionMode	The ServoPack will be controlled in position mode. A new command target must be calculated in the controller at a frequency suitable for the application. No acceleration or speed profiling is added by the system, except for the command filtering provided by pn 811/812, or pn 216/217 if using the MP2600iec.
2	VelocityTLMode	The ServoPack will be operated in velocity mode. The Torque input of the Y_DirectControl function block will serve as the torque Limit in this mode.
3	TorqueVLMODE	The ServoPack will be operated in torque mode. The Velocity input of the Y_DirectControl function block will serve as the velocity limit in this mode. If the velocity limit is reached, the servo will apply increased torque if necessary to keep the velocity within the limits.
4	PositionTrqFFMode	The ServoPack will be operated in position mode. The Torque input of the Y_DirectControl function block will serve as the torque feed forward value in this mode.
5	VelocityTrqFFMode	The ServoPack will be operated in velocity mode. The Torque input of the Y_DirectControl function block will serve as the torque feed forward value in this mode.

Y_TableType

0	Undefined	
1	MasterSlaveArray	This is the only TableType supported.